QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/482A(USAF)
18 June 1984
SUPERSEDING
MIL-M-38510/482(USAF)
16 March 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE

MONOLITHIC DIRECT MEMORY ACCESS CONTROLLER

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic N-channel, silicon gate direct memory access controller. One product assurance class and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device types. The device types shall be as follows:

Device type	Clock frequency	Circuit
01	4.0 MHz	Direct memory access controller
02	2.5 MHz	

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
Q	D-5 (40-lead, 9/16" x 2"), dual-in-line package

1.3 Absolute maximum ratings.

V_{CC} supply voltage range (V_{CC} - GND)	-0.3 V dc to +7 V dc
Voltage on any pin (referenced to ground)	-0.3 V dc to +7 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation	
T_C = -55°C	1.5 W
T_C = +125°C	1.0 W
Lead temperature (soldering - 5 seconds)	+270°C
Thermal resistance, junction-to-case (θ_{JC})	
Case Q	40°C/W
Junction temperature (T_J)	+170°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc (minimum) to 5.5 V dc (maximum)
Minimum high-level input voltage (V_{IH})	
Logic inputs	2.0 V dc
Clock input	V_{CC} -0.6 V dc
Maximum low-level input voltage (V_{IL})	
Logic inputs	0.8 V dc
Clock input	0.45 V dc
Frequency of operation:	
Device type 01	0.5 to 4.0 MHz
Device type 02	0.5 to 2.5 MHz
Case operating temperature range (T_C)	-55°C to +125°C
Clock rise time (t_r)	30 ns maximum
Clock fall time (t_f)	30 ns maximum

INDEX

Table		Page
I.	Electrical Performance Characteristics	3
II.	Electrical Test Requirements	8
III.	Group A inspection	13
IV.	Input/Output conditions	19
V.	Test Vectors	(see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification For

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

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TABLE I. Electrical performance characteristics.

DC characteristics

Test	Symbol	Conditions 4/ (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)	Figure no. 6/	Reference no. 5/	Limits		Unit
					Min	Max	
Clock input low voltage	V_{IL1}				-0.3	0.45	V
Clock input high voltage	V_{IH1}				V_{CC} -.6	5.5	V
Input low voltage	V_{IL2}				-0.3	0.8	V
Input high voltage	V_{IH2}				2.0	5.5	V
Output low voltage	V_{OL}	I_{OL} = 2.0 mA			--	0.4	V
Output high voltage	V_{OH}	I_{OH} = -250 µA			2.4	---	V
Power supply current	I_{CC}	Outputs open T_C = -55°C V_{CC} = 5.5 V			---	272	mA
Tristate output leakage, low current	I_{ZL}	V_{IN} = 0.4 V			-10	+10	µA
Tristate output leakage, high current	I_{ZH}	V_{IN} = 2.4 V			-10	+10	µA
Open drain output leakage, low current	I_{LOL}	V_{IN} = 0.4 V			-10	+10	µA
Open drain output leakage, high current	I_{LOH}	V_{IN} = 2.4 V			-10	+10	µA
Input low current (input and bidirectional)	I_{IL}	V_{IN} = 0.4 V			-10	+10	µA
Input high current (input and bidirectional)	I_{IH}	V_{IN} = 2.4 V			-10	+10	µA
Maximum frequency Ø1 Ø2	f_{MAX}				--- ---	4.0 2.5	MHz MHz
Clock capacitance	C_{I1}					40	pF
Input capacitance	C_{I2}					5	pF
Output capacitance, all outputs and data bus	C_O				---	10	pF

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.
A.C. characteristics

Test	Symbol	Conditions 4/ (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10%; unless otherwise specified)	Figure no. 6/	Reference no. 5/	Device types 01 Min	01 Max	02 Min	02 Max	Unit
Clock cycle time	t_{cyc}	T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10%, C_L = 50 pF ±10%; unless otherwise specified	8	1	250	2,000	400	2,000	ns
Clock pulse width High	t_{PWH1}		8	2	110	3/, 7/	170	3/, 7/	ns
Clock pulse width Low	t_{PWL1}		8	3	110	3/, 8/	170	3/, 8/	ns
Clock time Rise	t_{rc}			4	---	30	---	30	ns
Clock time Fall	t_{fc}			5	---	30	---	30	
$\overline{AREQ}$ pulse width High	t_{PWH2}		9	39	3/, 9/	---	3/, 9/	---	ns
$\overline{AREQ}$ pulse width Low	t_{PWL2}		8	40	3/, 10/	---	3/, 10/	---	
$\overline{WR}$ pulse width low	t_{PWL3}		8	52	3/, 10/	---	3/, 10/	---	ns
Data in to clock ↓ ($\overline{WR}$ or $\overline{MI}$)	t_{SZL1} t_{SZH1}		7	9	50	---	50	---	ns
$\overline{IET}$ ↓ to $\overline{IORD}$ ↓ setup ($\overline{INTA}$ cycle)	t_{SHL1}		7	12	140	---	140	---	ns
$\overline{MI}$ ↓ to clock ↑ setup	t_{SHL2}		7	16	90	---	210	---	ns
$\overline{MI}$ ↑ to clock ↑ setup	t_{SLH3}		7	17	-10	---	20	---	ns
$\overline{RD}$ to clock ↑ setup ($\overline{MI}$ cycle)	t_{SHL4} t_{SLH4}		7	18	115	---	240	---	ns
RDY to clock ↑ setup	t_{SLH5} t_{SHL5}		7	22	100	---	150	---	ns
Data in to clock ↑ setup (Read cycle when rising edge ends Read)	t_{SLH6} t_{SHL6}		8	31	35	---	50	---	ns
Data in to clock ↑ setup (Read cycle when falling edge ends Read)	t_{SLH7} t_{SHL7}		8	32	50	---	60	---	ns
$\overline{MI}$ ↓ to IEO ↓ delay 1/	t_{PHL9}		7	15	---	190	---	300	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

AC characteristics

Test	Symbol	Conditions 4/ (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)	Figure no. 6/	Reference no. 5/	Device type 01 Min	Device type 01 Max	Device type 02 Min	Device type 02 Max	Unit
Interrupt cause to $\overline{INT}$ ↑ delay 2/	t_{PHL10}	Figure 5 load	7	19	---	500	---	500	ns
$\overline{BAI}$ ↓ to $\overline{BAO}$ ↓ delay	t_{PHL11}		7	20	---	150	---	200	ns
$\overline{BAI}$ ↑ to $\overline{BAO}$ ↑ delay	t_{PHL12}		7	21	---	150	---	200	ns
Clock ↓ to address output delay	t_{PHL13} t_{PLH13}		8	23	---	110	---	145	ns
Clock ↑ to address float delay	t_{PHZ2} t_{PLZ2}		8	24	---	90	---	110	ns
$\overline{RD}$, $\overline{WR}$ ↑ to address stable delay	t_{PHL14} t_{PLH14}		8	27	3/, 11/	---	3/, 11/	---	ns
$\overline{RD}$, $\overline{WR}$ ↑ to address float delay	t_{PH73} t_{PL73}		8	28	3/, 12/	---	3/, 12/	---	ns
Clock ↓ to data out delay	t_{PHL15} t_{PLH15}		8	29	---	150	---	230	ns
Clock ↑ to data float delay (write cycle)	t_{PHZ4} t_{PLZ4}		8	30	---	90	---	90	ns
$\overline{WR}$ ↓ to data out delay	t_{PHL16} t_{PLH16}		8	35	3/, 13/		3/, 13/		ns
Clock ↓ to $\overline{MREQ}$ delay	t_{PHL17} t_{PLH17}		8	36 38	--	85		100	ns
Clock ↓ to $\overline{MREQ}$ ↑ delay	t_{PLH18}		8	37	---	85	---	100	ns
$\overline{WAIT}$ to clock ↓ setup	t_{SLH8} t_{SHL8}		8	53	70	---	70	---	ns
$\overline{IORQ}$, $\overline{WR}$, $\overline{CE}$ ↓ to clock ↓ setup time	t_{SHL9} t_{SLH9}		7	7	145	--	200	---	ns
Hold time for any specified setup time	t_{HHL1} t_{HLH1}		7, 8	6	0	---	0	--	ns
Address to $\overline{MREQ}$ ↓ delay (memory cycle)	t_{PHL1} t_{PLH1}		8	26	3/, 14/		3/, 14/	---	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.
AC characteristics

Test	Symbol	Conditions 4/ (T_C = -55°C to +125°C, V_{CC} = 5.0 V ±10% unless otherwise specified)	Figure no. 6/	Reference no. 5/	Device type 01 Min	01 Max	02 Min	02 Max	Unit
Address stable prior to IORQ, RD, WR ↓ delay (I/O cycle)	t_{PHL7} t_{PLH7}		8	26	3/, 15/	---	3/, 15/		ns
Data out to WR ↓ delay (memory cycle)	t_{PHL3} t_{PLH3}		8	32	3/, 15/	---	3/, 16/	---	ns
Data out to WR ↓ delay (I/O cycle)	t_{PHL4} t_{PLH4}		8	34	100	---	100	---	ns
RD ↓ to data output delay	t_{PHL5} t_{PLH5}		7	9	---	380	---	500	ns
IORQ ↓ to data out delay (INTA cycle)	t_{PHL6} t_{PLH6}		7	10	---	160	---	340	ns
RD ↑ to data float delay (output buffer disable)	t_{PHZ1} t_{PLZ1}		7	11	---	110	---	160	ns
TET ↑ to TEO ↑ delay	t_{PLH7}		7	13	-	160		210	ns
TET ↓ to TEO ↓ delay	t_{PHL8}		7	14	---	130	---	190	ns
Clock ↑ to IORQ delay	t_{PHL19} t_{PLH19}		8	41 42	--- ---	75 85	--- ---	90 100	ns
Clock ↑ to IORQ delay	t_{PLH20}		8	43	---	65	---	110	ns
Clock ↑ to RD delay	t_{PHL21} t_{PLH21}		8 8	44 45	---	85	-	100	ns
Clock ↓ to RD delay	t_{PHL22} t_{PLH22}		8 8	46 47	-- --	95 85	--- -	130 110	ns
Clock ↑ to WR delay	t_{PHL23} t_{PLH23}		8 8	48 50	--- ---	65 80	--- ---	80 100	ns
Clock ↓ to WR delay	t_{PHL24} t_{PLH24}		8 8	49 51	--- ---	80 80	--- ---	90 110	ns
Clock ↑ to BUSRQ delay	t_{PHL25} t_{PLH25}	Figure 5 load	8	54	---	100	---	100	ns
Clock ↑ to IORQ, MREQ, RD, WR float delay	t_{PLZ5} t_{PHZ5}		8	55	--	90	--	100	ns

See footnotes at end of table.

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1/ Interrupt just prior to M1 ↑ .

2/ $\overline{INT}$ generated only when DMA is inactive.

3/ Timing equations assume standard CPU CLK used. They do not necessarily apply to variable cycle timing.

4/ Loads for output pins not listed under condition are standard output loads (see figure 4).

5/ The reference number refers to the position where the parameter being tested appears on figures 7 and 8.

6/ Figures 7 and 8 represent the electrical performance characteristics associated with the ac limits shown in table I.

7/ $t_{cyc} - t_{PWL1} - t_{rc} - t_{fc}$

8/ $t_{cyc} - t_{PWH1} - t_{rc} - t_{fc}$

9/ $t_{PWH1} + t_{fc} - A$

10/ $t_{cyc} - B$

11/ $t_{PWL1} + t_{rc} - C$

12/ $t_{PWL1} + t_{rc} - D$

13/ $t_{PWL1} + t_{rc} - E$

14/ $t_{PWH1} + t_{fc} - 75$

15/ $t_{cyc} - E$

16/ $t_{cyc} - F$

	01	02
A	20	30
B	30	40
C	50	40
D	45	60
E	70	80
F	170	210

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3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.4.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and on figures 7 and 8, unless otherwise specified, apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. Electrical test requirements shall be as specified in tables III and IV, and on figures 7 and 8. The subgroups of table III which constitute the minimum electrical test requirements for screening and quality conformance by device class are specified in table II.

3.6 Register definitions. Register definitions shall be as specified in table V.

3.7 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in a microcircuit group number 107. (See MIL-M-38510, appendix E).

3.9 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line, not necessarily the line producing the device types described herein.

3.10 Certification. Certification in accordance with MIL-M-38510 is not required for these devices.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)
	Class B devices
Interim electrical parameters (method 5004)	1, 7
Final electrical test parameters (method 5004)	*1, 2, 3, *7, 8, 9, 10, 11
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11, 12
Group B zap test	See 4.5.2
Group C end-point electrical parameters (method 5005)	1, 2, 3, 7, 8
Group D end-point electrical parameters (method 5005)	1, 2, 3, 7, 8

*PDA applies to subgroups 1 and 7 (see 4.2d).

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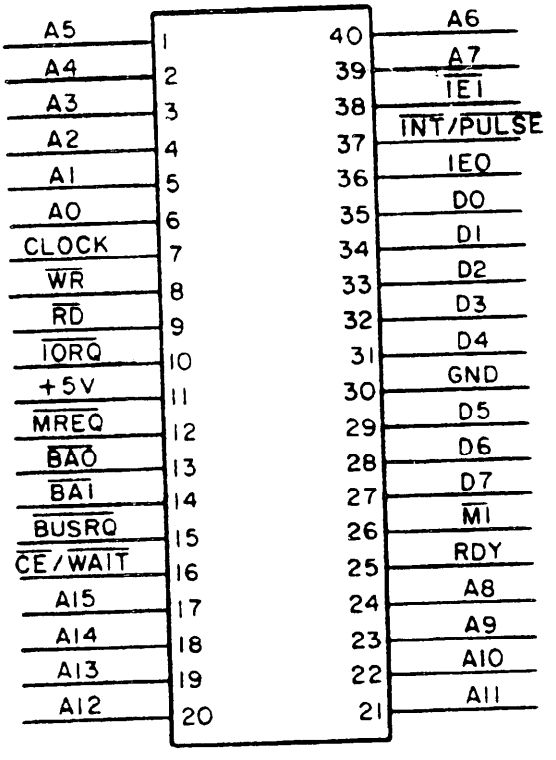


FIGURE 1. Terminal connections.

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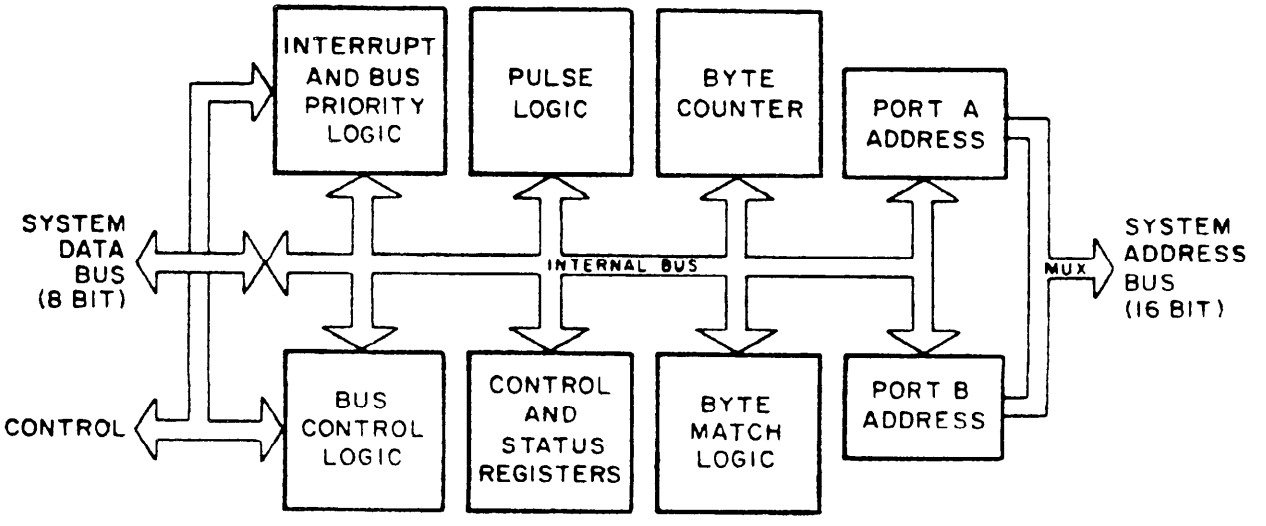
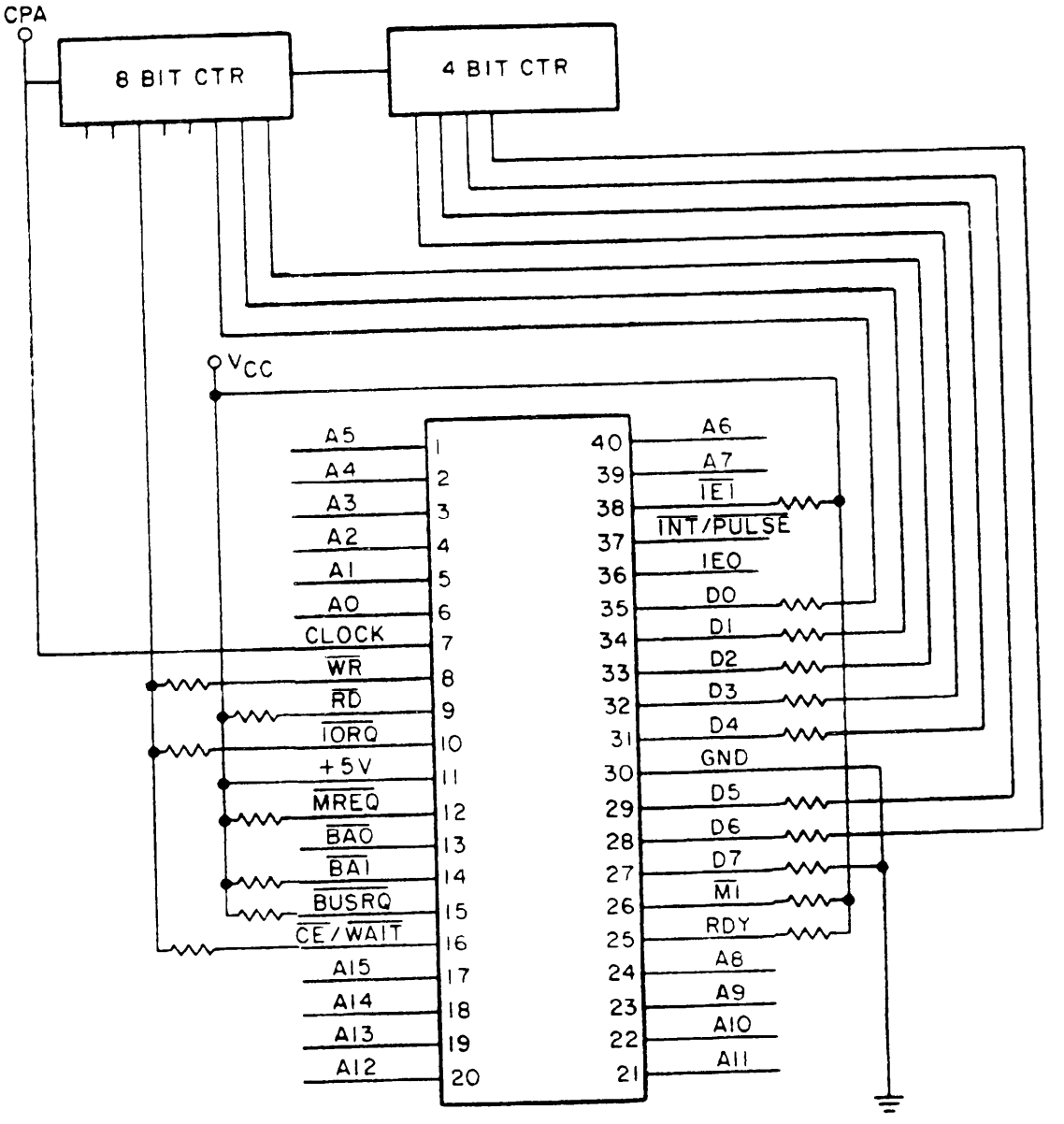


FIGURE 2. Functional block diagram.

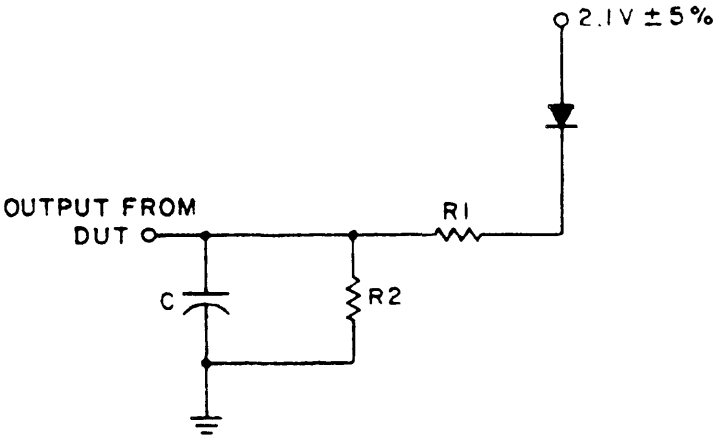
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NOTES:
1. CPA - 100 kHz ±10%
   50% duty cycle
2. V_{CC} shall be no less than 4.5 V at device terminal.
   V_{CC} shall be no more than 5.5 V at device terminal.
3. V_{IL_1} shall be no more than 0.5 V at device terminal.
   V_{IH_1} shall be no less than 3.9 V at device terminal.
4. All resistors to be 1 kΩ ±10%.

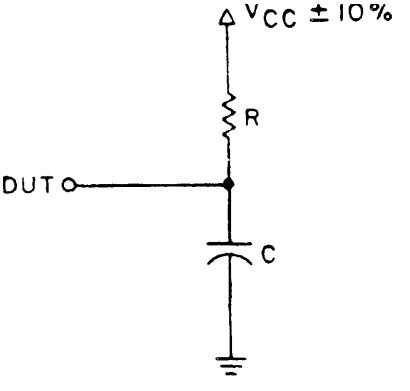
FIGURE 3. Burn-in and life test circuit.

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NOTES:
1. C = 50 pF ±5% including tester, probe, wiring and stray capacitance.
2. R1 = 510Ω ±5%, R2 = 9.53 kΩ ±1%.
3. The diode is 1N3064 or equivalent.

FIGURE 4. Output circuit for standard outputs for ac and functional testing, both device types.



NOTES:
1. C = 50 pF ±5%, including tester, wiring and stray capacitance.
2. R = 2.2 kΩ ±5%.
3. Open drain outputs are $\overline{BUSR/Q}$, $\overline{INT/Pulse}$

FIGURE 5. Output load circuit for open drain outputs for ac and functional testing, both device types.

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See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated are open).

Test no.	21 A_{11}	22 A_{10}	23 A_9	24 A_8	25 RDY	26 RT	27 D_7	28 D_6	29 D_5	30 GND	31 D_4	32 D_3	33 D_2	34 D_1	35 D_0	36 IEO	37 $\overline{IWT}/PISE$	38 IEI	39 A_7	40 A_6	Measured terminal	Test limits Min	Test limits Max	Unit
																					V_{CC}		200	µA
																						-10	10	µA
1					2.0 V	2.0 V				GND								2.0 V						
2																					A_5			
3																					A_4			
4																					A_3			
5																					A_2			
6																					A_1			
7																					A_0			
8																					WRITE			
9																					A_{15}			
10																					A_{14}			
11																					A_{13}			
12																					A_{12}			
13	0.4 V																				A_{11}			
14		0.4 V																			A_{10}			
15			0.4 V																		A_9			
16				0.4 V																	A_8			
17																					A_7			
18																				2.4 V	A_6			
19																					A_5			
20																					A_4			
21																					A_3			
22																					A_2			
23																					A_1			
24																					A_0			
25																					WRITE			
26																					A_{15}			
27																					A_{14}			
28																					A_{13}			
29																					A_{12}			
30	2.4 V																				A_{11}			
31		2.4 V																			A_{10}			
32			2.4 V																		A_9			
33				2.4 V																	A_8			
34																					A_7			
35																				2.4 V	A_6			
36																					CLK			
37																					$\overline{WR}$			
38																					$\overline{RD}$			
39																					$\overline{IOWR}$			
40																					$\overline{BAI}$			
41																					CE/BAIT			
42					0.4 V	0.4 V															RDY			
43					2.0 V	2.0 V															$\overline{BT}$			
44							0.4 V														D_7			
45								0.4 V													D_6			
46									0.4 V												D_5			
47											0.4 V										D_4			
48												0.4 V									D_3			
49													0.4 V								D_2			
50														0.4 V							D_1			
51															0.4 V						D_0			
52																				0.4 V	IEI			
53																					CLK			
54																					$\overline{WR}$			
55																					$\overline{RD}$			
56																					$\overline{IOWR}$			
57																					$\overline{BAI}$			
58																					CE/BAIT			
59					2.4 V	2.4 V															RDY			
60					2.0 V	2.0 V															$\overline{BT}$			
61							2.4 V														D_7			
62								2.4 V													D_6			
63									2.4 V												D_5			
64											2.4 V										D_4			
65												2.4 V									D_3			
66													2.4 V								D_2			
67														2.4 V							D_1			
68															2.4 V						D_0			
69																					IEO			

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TABLE III. Group A inspection for device types 01 and 12 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no	A_3	A_4	A_5	A_6	A_7	A_8	CLK	QR	QS	V_{CC}	QRFS	ERR	ERR STROBE	A_{12}	A_{13}	A_{14}
1 T_C = +25°C	I_{CCL}		71							4.5 V	2.0 V	2.0 V	5.5 V						
	I_{CCH}		72							4.5 V	2.0 V	2.0 V	5.5 V						
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and the limit for test 71 is N/A																		
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and the limit for test 71 is N/A																		
4 T_C = +25°C	V_{OL}		73	GND	GND		GND	GND		1/ GND	GND	GND	GND	GND	GND	GND	GND	GND	GND
	V_{OH}		74	GND	GND		GND	GND		1/ GND	GND	GND	GND	GND	GND	GND	GND	GND	GND
			75																
			76																
			77																
			78																
			79																
			80																
			81	1/ GND															
			82		1/ GND														
			83					1/ GND											
			84						1/ GND										
			85							1/ GND									
			86								1/ GND								
			87									1/ GND							
			88										1/ GND						
			89											1/ GND					
			90												1/ GND				
			91													1/ GND			
			92														1/ GND		
			93															1/ GND	
			94																1/ GND
			95																
			96																
			97																
			98																
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			103																
			104																
			105																
			106																
			107																
			108																
			109																
			110																
			111																
7 T_C = +25°C	12/ 37/ 13/	3014	See 5.1 (Table V)																
8	12/ 37/ 13/		Same tests, terminal conditions and limits as for subgroup 7 except T_C = +125°C and T_C = -55°C.																
9 T_C = +25°C	12/ 37/ 13/																		
10 T_C = +25°C	12/ 37/ 13/																		
11 T_C = +25°C	12/ 37/ 13/																		

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TABLE III. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated may be open).

Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
70	A_{11}	A_{10}	A_9	A_8	RDY	RT	D_7	D_6	D_5	GND	D_4	D_3	D_2	D_1	D_0	IE1	INT/PULSE	IE1	A_7	A_6				
71					2.0 V	2.0 V				GND							0.8 V	2.0 V						
72																					(IORQ) (WR/PULSE)			
73																								
74	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND				
75																					(M1) (IORQ) CE/WAIT RD1 WR (INT/PULSE)			
76																								
77																								
78					1/GND																			
79																								
80						1/GND																		
81																					A_5			
82																					A_4			
83																					A_3			
84																					A_2			
85																					A_1			
86																					A_0			
87																					RD			
88																					(IORQ) WR/TEST			
89																					BUSRQ			
90																					A_{15}			
91																					A_{14}			
92																					A_{13}			
93																					A_{12}			
94																					A_{11}			
95																					A_{10}			
96																					A_9			
97																					A_8			
98																					D_7			
99																					D_6			
100																					D_5			
101																					D_4			
102																					D_3			
103																					D_2			
104																					D_1			
105																					D_0			
106																					IE1			
107																					A_7			
108																					A_6			
109																								
110																								
111																								

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TABLE IIIA. Group A inspection.

Symbol	Test no. 8/	Terminal(s)	Figure number 16/	Reference number 15/	Subgroup 12 limits +25°C, +125°C				Unit
					Device types				
					01		02		
					Minimum	Maximum	Minimum	Maximum	
t_{PHL1} t_{PLH1}	112	$\overline{MREQ}$	8	25	9/		9/		ns
t_{PHL2} t_{PLH2}	113	$\overline{IORQ}$, $\overline{RD}$, $\overline{WR}$	"	26	10/		10/		"
t_{PHL3} t_{PLH3}	114	$\overline{WR}$	"	33	11/		11/		"
t_{PHL4} t_{PLH4}	115	$\overline{WR}$	"	34	100		100		"
t_{PLH5} t_{PHL5}	116	$\overline{RD}$	7	8		380		500	"
t_{PHL6} t_{PLH6}	117	$\overline{IORQ}$	"	10		160		340	"
t_{PLH7}	118	$\overline{IEO}$	"	13		160		210	"
t_{PHL8}	119	$\overline{IEO}$	"	14		130		190	"
t_{PHL9}	120	$\overline{IEO}$	"	15		190		300	"
t_{PHL10}	121	$\overline{INT}$	"	19		500		500	"
t_{PLH11}	122	$\overline{BAO}$	"	20		150		200	"
t_{PHL12}	123	$\overline{BAO}$	"	21		150		200	"
t_{PHL13} t_{PLH13}	124	A_0-A_{15}	8	23		110		145	"
t_{PHL14} t_{PLH14}	125	$\overline{RD}$, $\overline{WR}$	"	27		12/		12/	"
t_{PHL15} t_{PLH15}	126	D_0-D_7	"	29		90		90	"
t_{PHL16} t_{PLH16}	127	D_0-D_7	"	35	13/		13/		"
t_{PHL17} t_{PLH17}	128	$\overline{MREQ}$	"	36 38		85		100	"
t_{PHL18}	129	$\overline{MREQ}$	"	37		85		100	"
t_{PHL19}	130	$\overline{IORQ}$	"	41		75		90	"
t_{PLH19}	131	$\overline{IORQ}$	"	42		85		100	"
t_{PHL20}	132	$\overline{IORQ}$	"	43		85		110	"
t_{PLH21} t_{PHL21}	133	$\overline{RD}$	"	44 46		85		100	"
t_{PHL22}	134	$\overline{RD}$	"	45		95		130	"

See footnotes at end of table.

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TABLE IIIA. Group A inspection - Continued.

Symbol	Test no. 8/	Terminal(s)	Figure number 16/	Reference number 15/	Subgroup 12 limits +25°C, +125°C				Unit
					Device types				
					01		02		
					Minimum	Maximum	Minimum	Maximum	
t_{PLH22}	135	$\overline{RD}$	8	47		85		110	ns
t_{PHL23}	136	$\overline{WR}$	"	48		65		80	"
t_{PLH23}	137	$\overline{WR}$	"	50		80		100	"
t_{PHL24}	138	$\overline{WR}$	"	49		80		90	"
t_{PLH24}	139	$\overline{WR}$	"	51		80		100	"
t_{PHL25} t_{PLH25}	140	$\overline{BUSRQ}$	"	55		100		100	"
t_{PHZ1} t_{PLZ1}	141	D_0-D_7	7	11		110		160	"
t_{PHZ2} t_{PLZ2}	142	D_0-D_7	8	24		90		110	"
t_{PHZ3} t_{PLZ3}	143	A_0-A_{15}	"	28		147		147	"
t_{PHZ4} t_{PLZ4}	144	D_0-D_7	"	30		90		90	"
t_{PHZ5} t_{PLZ5}	145	$\overline{IORQ}$, $\overline{MREQ}$, $\overline{RD}$, $\overline{WR}$	"	55		80		100	"

1/ The capacitance measurement shall be made between the indicated terminal and ground at a frequency of 1 MHz. The dc bias of the measurement instrument shall be less than ±0.1 V. The ac amplitude shall be less than 50 mV rms.
2/ For functional tests the input/output conditions A, B, C, and D appear in table IV.
3/ Table V functional tests shall be repeated for input/output conditions A through D of table IV.
4/ All subgroups 9, 10, and 11 are run concurrently with subgroups 7 and 8.
5/ Figures 7 and 8 represents the timing diagrams associated with subgroups 9, 10, and 11.
6/ V_{OH} and V_{OL} are measured as part of subgroups 7 and 8.
7/ Where propagation delay tests for subgroup 12 indicate a group of pins, and where two or more propagation delays are specified for a test, only the reading for the pin with the worst delay requires recording. In some cases, this will be a minimum reading while in others it will be a maximum reading as indicated in the table.
8/ Table V functional tests shall be repeated for input/output test conditions A through D of table IV when making measurements for tests 117 through 145.

9/ $t_{PWH1} + t_{rc}$ -75
10/ t_{cyc} - A
11/ t_{cyc} - B
12/ $t_{PWL1} + t_{rc}$ - C
13/ $t_{PWL1} + t_{rc}$ - D
14/ $t_{PWL1} + t_{rc}$ - E

	01	02
A	70	80
B	170	210
C	50	40
D	70	80
E	45	60

15/ The reference number refers to the position where the parameter being tested appears on figures 7 and 8.
16/ Figures 7 and 8 represent the electrical performance characteristics associated with the limits shown in table I.

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TABLE IV. Input/output conditions for table III, device type 01.

Symbol	Reference number 12/	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
V_{CC}	---	V_{CC}	5.5	4.5	4.5	4.5	V
V_{IH2}	---	Logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}	---	Logic inputs	0.8	0.8	0.8	0.8	V
V_{IH1}	---	CLK	4.9	3.9	4.9	3.9	V
V_{IL1}	---	CLK	0.45	0.45	0.45	0.45	V
V_{OH}	---	Outputs	2.4	2.4	2.4	2.4	V
V_{OL}	---	Outputs	0.4	0.4	0.4	0.4	V
t_{cyc}	1	CLK	250	250	2000	2000	ns
t_{PWL1}	3	CLK	110	110	110		ns
t_{PWH1}	2	CLK				110	ns
t_{SHL9}	7	$\overline{IORQ}$, $\overline{WR}$, $\overline{CE}$	145	145	2,800	2,800	ns
t_{SZL1} t_{SZH1}	9	D_0-D_7	50	50	50	50	ns
t_{SHL1}	12	$\overline{IEI}$	140	140	140	140	ns
t_{SHL2}	16	$\overline{M1}$	90	90	2,100	2,100	ns
t_{SLH3}	17	$\overline{M1}$	-10	-10	200	200	ns
t_{SHL4} t_{SLH4}	18	$\overline{RD}$	115	115	2,400	2,400	ns
t_{SLH5} t_{SHL5}	22	RDY	100	100	1,500	1,500	ns
t_{SLH7} t_{SHL7}	32	D_0-D_7	50	50	600	600	ns
t_{SLH8} t_{SHL8}	53	$\overline{WAIT}$	70	70	70	70	ns
t_{PLH6} t_{PHL6}	10	D_0-D_7	160	160	3,400	3,400	ns
t_{PLH7}	13	IEO	160	160	2,100	2,100	ns
t_{PHL8}	14	IEO	130	130	1,900	1,900	ns
t_{PHL9}	15	IEO	190	190	3,000	3,000	ns
t_{PLH11}	20	$\overline{BAO}$	150	150	2,000	2,000	ns
t_{PHL12}	21	$\overline{BAO}$	150	150	2,000	2,000	ns
t_{PLH13} t_{PHL13}	23	A_0-A_{15}	110	110	1,450	1,450	ns

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III, device type 01- Continued.

Symbol	Reference number 12/	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
t_{PLH15} t_{PHL15}	29	D_0-D_7	150	150	2,300	2,300	ns
t_{PHL17} t_{PLH17}	36 38	$\overline{MREQ}$	85	85	1,000	1,000	ns
t_{PLH18}	37	$\overline{MREQ}$	85	85	1,000	1,000	ns
t_{PHL19}	41	$\overline{IORQ}$	75	75	900	900	ns
t_{PLH19}	42	$\overline{IORQ}$	85	85	1,000	1,000	ns
t_{PLH20}	43	$\overline{IORQ}$	85	85	1,100	1,100	ns
t_{PHL21}	44	$\overline{RD}$	85	85	1,000	1,000	ns
t_{PLH21}	46	---	---	---	---	---	---
t_{PHL22}	45	$\overline{RD}$	95	95	1,300	1,300	ns
t_{PLH22}	47	$\overline{RD}$	85	85	1,100	1,100	ns
t_{PHL23}	48	$\overline{WR}$	65	65	800	800	ns
t_{PLH23}	50	$\overline{WR}$	80	80	1,000	1,000	ns
t_{PHL24}	49	$\overline{WR}$	80	80	900	900	ns
t_{PLH24}	51	$\overline{WR}$	80	80	1,000	1,000	ns
t_{PLH25} t_{PHL25}	54	BUSRQ	100	100	100	100	ns
t_{PLH5} t_{PHL5}	8	$\overline{RD}$	380	380	380	380	ns
t_{PLZ1} t_{PHZ1}	11	$\overline{RD}$	110	110	110	110	ns
t_{PHL10}	19	$\overline{INT}$	500	500	500	500	ns
t_{PLZ2} t_{PHZ2}	24	A_8-A_{15}	90	90	90	90	ns
t_{PHL14} t_{PLH14}	27	$\overline{RD}$, $\overline{WR}$	1/	1/	1/	1/	ns
t_{PLZ3} t_{PHZ3}	28	$\overline{RD}$, $\overline{WR}$	2/	2/	2/	2/	ns
t_{PLZ4} t_{PHZ4}	30	D_0-D_8	90	90	90	90	ns
t_{PLH16} t_{PHL16}	35	$\overline{WR}$	3/	3/	3/	3/	ns
t_{PLZ5} t_{PHZ5}	55	$\overline{IORQ}$, $\overline{MREQ}$, $\overline{RD}$, $\overline{WR}$	80	80	80	80	ns

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III, device type 01- Continued.

Symbol	Reference number 12/	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
t_{SLH2} t_{SHL2}	26	$\overline{IORQ}$, $\overline{RD}$, $\overline{RW}$	7/	7/	7/	7/	ns
t_{SLH6} t_{SHL6}	31	D_0-D_7	35	35	35	35	ns
t_{PHL1} t_{PLH1}	25	$\overline{MREQ}$	8/	8/	8/	8/	ns
t_{PHL4} t_{PLH4}	34	$\overline{WR}$	100	100	100	100	ns
t_{PHL3} t_{PLH3}	33	$\overline{WR}$	9/	9/	9/	9/	ns
t_{HLH1} t_{HHL1}	6	Hold time for any specified setup time	0	0	0	0	ns

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III, device type 02.

Symbol	Reference number 12/	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
V_{CC}	---	V_{CC}	5.5	4.5	4.5	4.5	V
V_{IH2}	---	Logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}	---	Logic inputs	0.8	0.8	0.8	0.8	V
V_{IH1}	---	CLK	4.9	3.9	4.9	3.9	V
V_{IL1}	---	CLK	0.45	0.45	0.45	0.45	V
V_{OH}	---	Outputs	2.4	2.4	2.4	2.4	V
V_{OL}	---	Outputs	0.4	0.4	0.4	0.4	V
t_{cyc}	1	CLK	400	400	2000	2000	ns
t_{PWL1}	3	CLK	170	170	170		ns
t_{PWH1}	2	CLK				170	ns
t_{SHL9}	7	$\overline{IORQ}.\overline{WR}.CE$	280	280	2,800	2,800	ns
t_{SZL1} t_{SZH1}	9	D_0-D_7	50	50	50	50	ns
t_{SHL1}	12	IEI	140	140	140	140	ns
t_{SHL2}	16	$\overline{M1}$	210	210	2,100	2,100	ns
t_{SLH3}	17	$\overline{M1}$	20	20	200	200	ns
t_{SHL4} t_{SLH4}	18	$\overline{RD}$	240	240	2,400	2,400	ns
t_{SLH5} t_{SHL5}	22	RDY	150	150	1,500	1,500	ns
t_{SLH7} t_{SHL7}	32	D_0-D_7	60	60	600	600	ns
t_{SLH8} t_{SHL8}	53	$\overline{WAIT}$	70	70	70	70	ns
t_{PLH6} t_{PHL6}	10	D_0-D_7	340	340	3,400	3,400	ns
t_{PLH7}	13	IEO	210	210	2,100	2,100	ns
t_{PHL8}	14	IEO	190	190	1,900	1,900	ns
t_{PHL9}	15	IEO	300	300	3,000	3,000	ns
t_{PLH11}	20	$\overline{BAO}$	200	200	2,000	2,000	ns
t_{PHL12}	21	$\overline{BAO}$	200	200	2,000	2,000	ns
t_{PLH13} t_{PHL13}	23	A_0-A_{15}	145	145	1,450	1,450	ns

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III, device type 02- Continued.

Symbol	Reference number 12/	Terminal(s)	Input/output conditions A	B	C	D	Unit
t_{PLH15} t_{PHL15}	29	D_0-D_7	230	230	2,300	2,300	ns
t_{PHL17} t_{PLH17}	36 38	$\overline{MREQ}$	100	100	1,000	1,000	ns
t_{PLH18}	37	$\overline{MREQ}$	100	100	1,000	1,000	ns
t_{PHL19}	41	$\overline{IORQ}$	90	90	900	900	ns
t_{PLH19}	42	$\overline{IORQ}$	100	100	1,000	1,000	ns
t_{PLH20}	43	$\overline{IORQ}$	110	100	1,100	1,100	ns
t_{PHL21}	44	$\overline{RD}$	100	100	1,000	1,000	ns
t_{PLH21}	46	---	---	---	---	---	ns
t_{PHL22}	45	$\overline{RD}$	130	130	1,300	1,300	ns
t_{PLH22}	47	$\overline{RD}$	110	110	1,100	1,100	ns
t_{PHL23}	48	$\overline{WR}$	80	80	800	800	ns
t_{PLH23}	50	$\overline{WR}$	100	100	1,000	1,000	ns
t_{PHL24}	49	$\overline{WR}$	90	90	900	900	ns
t_{PLH24}	51	$\overline{WR}$	100	100	1,000	1,000	ns
t_{PLH25} t_{PHL25}	54	$\overline{BUSRQ}$	100	100	100	100	ns
t_{PLH5} t_{PHL5}	8	$\overline{RD}$	500	500	500	500	ns
t_{PLZ1} t_{PHZ1}	11	$\overline{RD}$	160	160	160	160	ns
t_{PHL10}	19	$\overline{INT}$	500	500	500	500	ns
t_{PHZ2} t_{PLZ2}	24	A_8-A_{15}	110	110	110	110	ns
t_{PHL14} t_{PLH14}	27	$\overline{RD}$, $\overline{WR}$	4/	4/	4/	4/	ns
t_{PLZ3} t_{PHL3}	28	$\overline{RD}$, $\overline{WR}$	5/	5/	5/	5/	ns
t_{PLZ4} t_{PHZ4}	30	D_0-D_8	90	90	90	90	ns
t_{PLH16} t_{PHL16}	35	$\overline{WR}$	6/	6/	6/	6/	ns
t_{PLZ5} t_{PHZ5}	55	$\overline{IORQ}$, $\overline{MREQ}$, $\overline{RD}$, $\overline{WR}$	100	100	100	100	ns

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III, device type 02- Continued.

Symbol	Reference number 12/	Terminal(s)	Input/output conditions A	B	C	D	Unit
t_{SLH2} t_{SHL2}	26	$\overline{IORQ}$, $\overline{RD}$, $\overline{RW}$	10/	10/	10/	10/	ns
t_{SLH6} t_{SHL6}	31	D_0-D_7	50	50	50	50	ns
t_{PHL1} t_{PLH1}	25	$\overline{MREQ}$	8/	8/	8/	8/	ns
t_{PHL4} t_{PLH4}	34	$\overline{WR}$	100	100	100	100	ns
t_{PHL3} t_{PLH3}	33	$\overline{WR}$	11/	11/	11/	11/	ns
t_{HLH1} t_{HHL1}	6	Hold time for any specified setup time	0	0	0	0	ns

1/ $t_{PWL1} + t_{rc}$ -50 ns

2/ $t_{PWL1} + t_{rc}$ -45 ns

3/ $t_{PWL1} + t_{rc}$ -70 ns

4/ $t_{PWL1} + t_{rc}$ -40 ns

5/ $t_{PWL1} + t_{rc}$ -60 ns

6/ $t_{PWL1} + t_{rc}$ -80 ns

7/ t_{cyc} - 70

8/ $t_{PWH1} + t_{fc}$ - 70

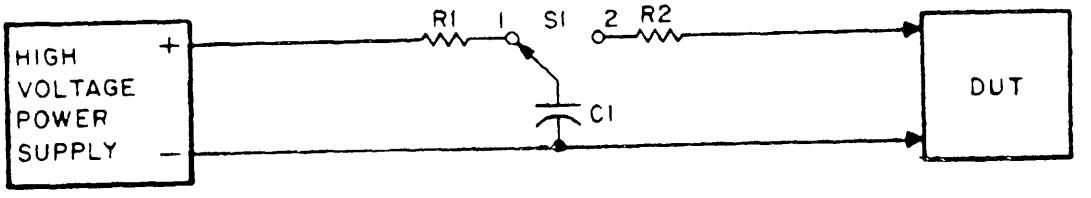
9/ t_{cyc} - 170

10/ t_{cyc} - 80

11/ t_{cyc} - 210

12/ The reference number refers to the position where the parameter being tested appears on figures 7 and 8.

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NOTES:
1. V_{ZAP} = 400 V minimum, measured at the DUT input terminal.
2. 1 MΩ ≤ R1 ≤ 5 MΩ.
3. R2 = 1.5 kΩ ±5%.
4. C1 = 100 pF ±10%.
5. S1 = Hg-wetted "bounceless" relay.

FIGURE 6. High voltage (V_{ZAP}) test circuits.

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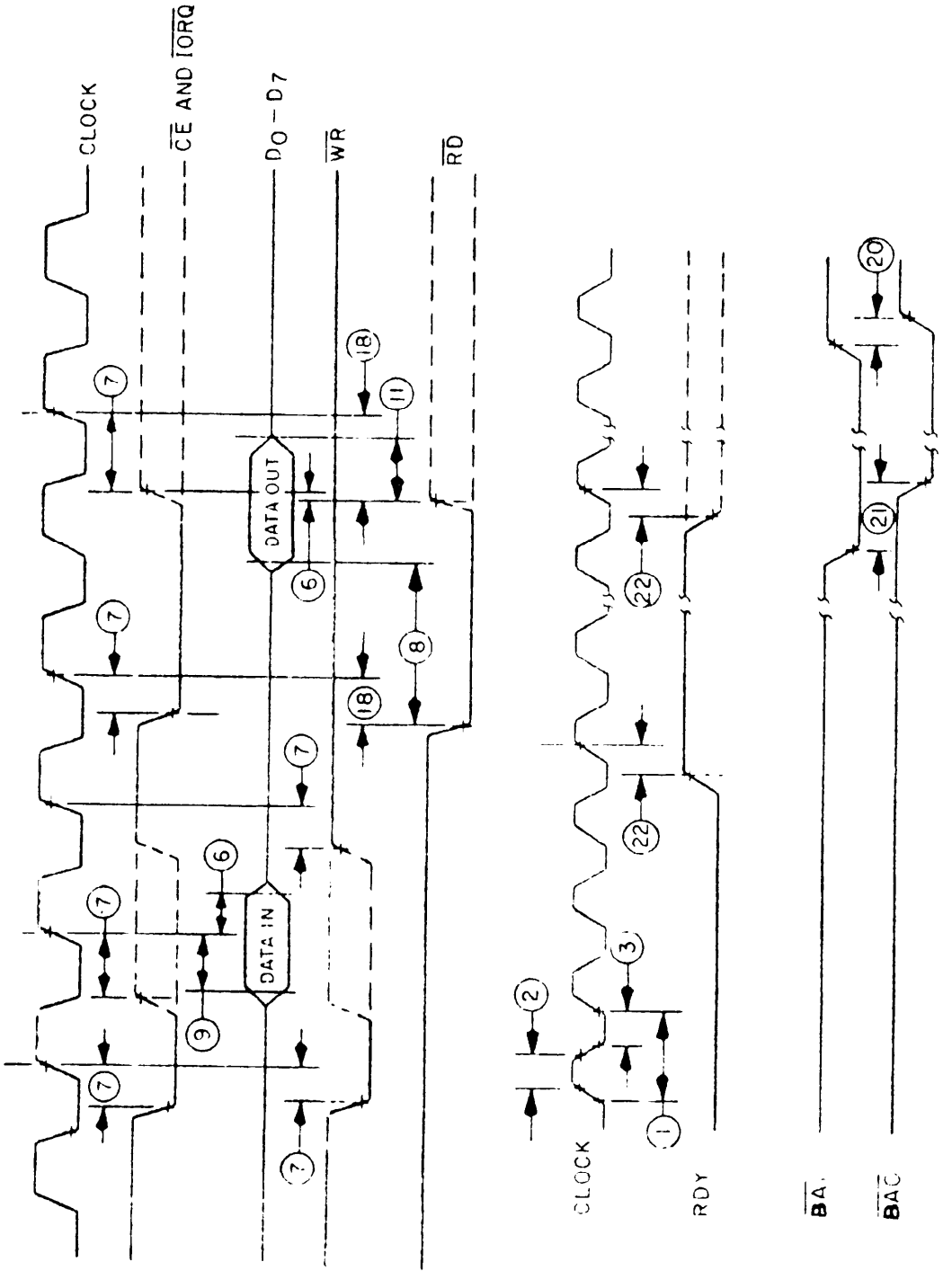
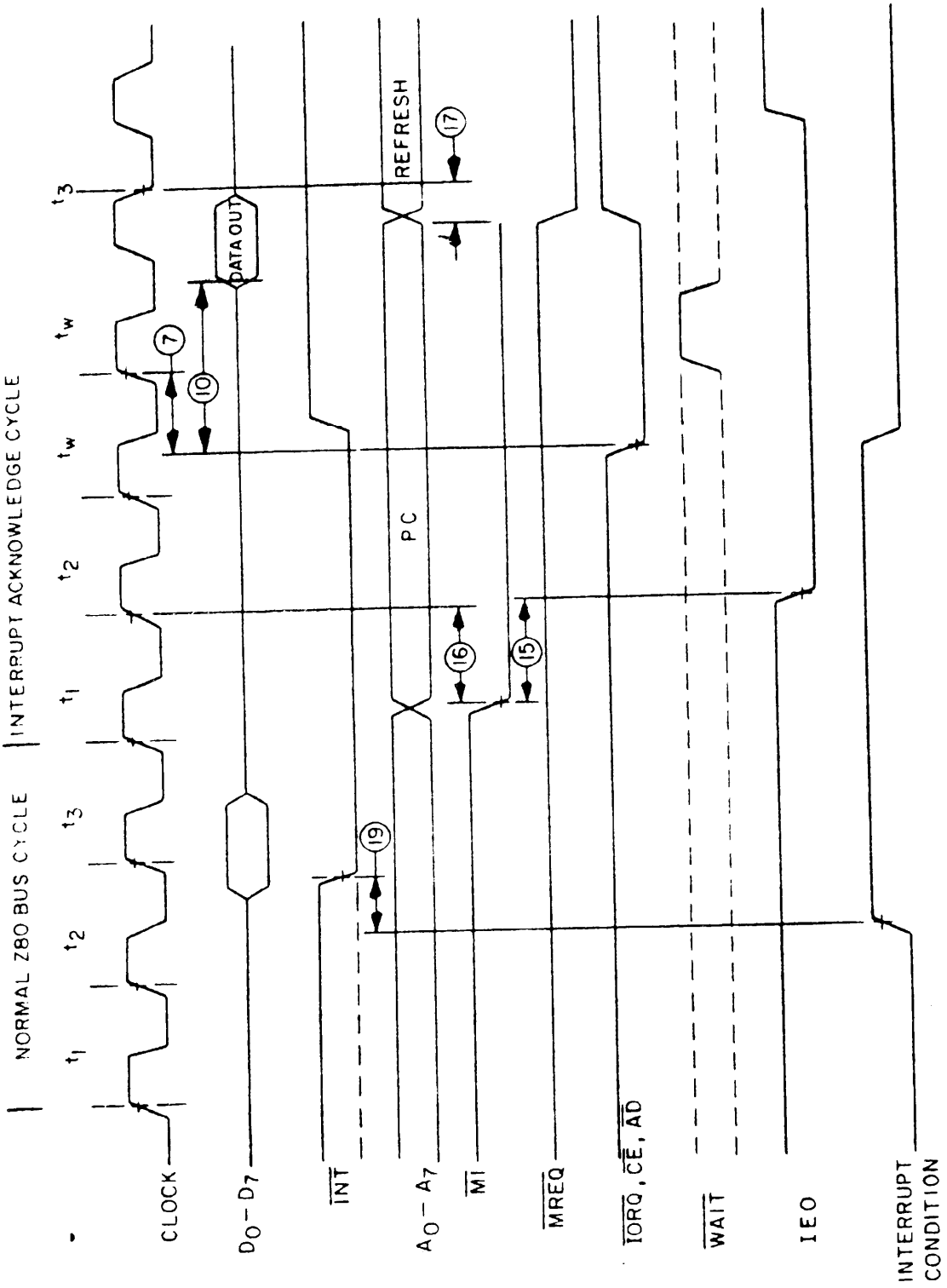


FIGURE 7. Inactive state ac characteristics.

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NOTE: Interrupt conditions can include end of block, byte match, or active ray line depending on programming.

FIGURE 7. Inactive state ac characteristics - Continued.

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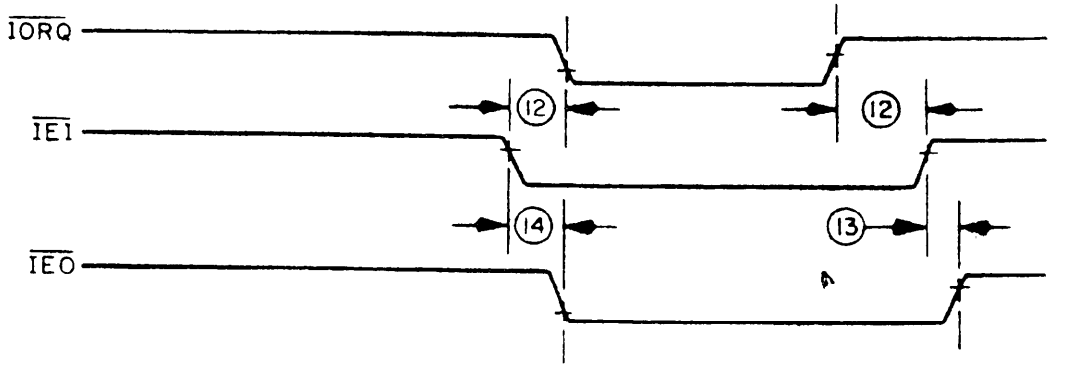


FIGURE 7. Inactive state ac characteristics - Continued.

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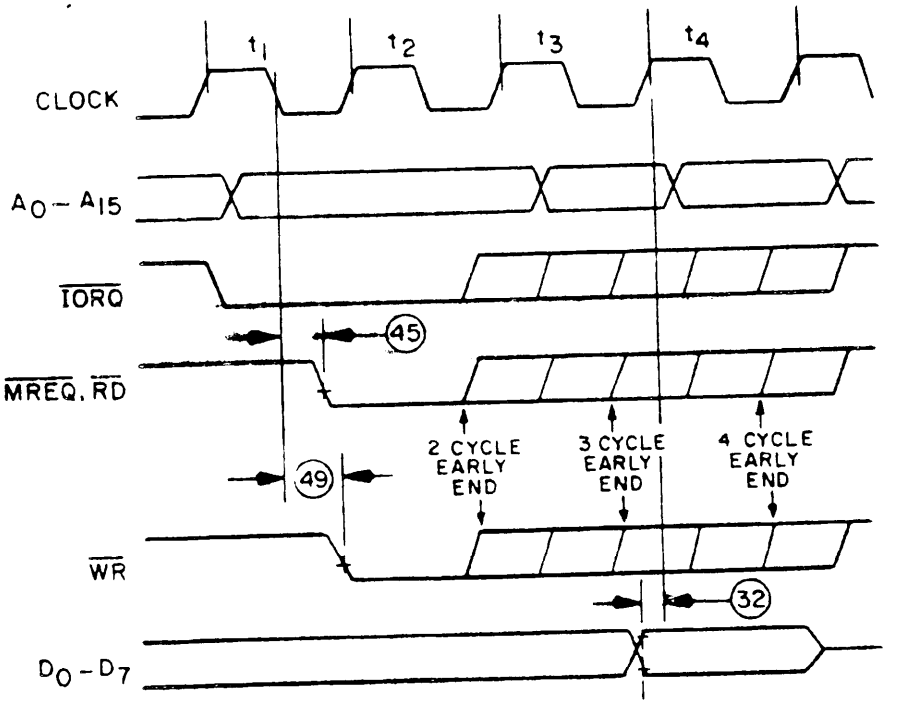


FIGURE 8. Variable E cycle and edge timing.

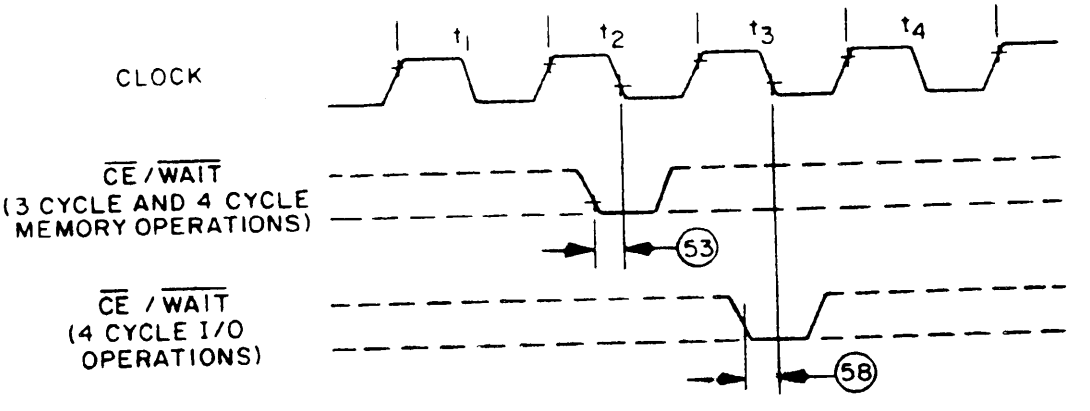


FIGURE 8. Wait line sampling in variable cycle timing.

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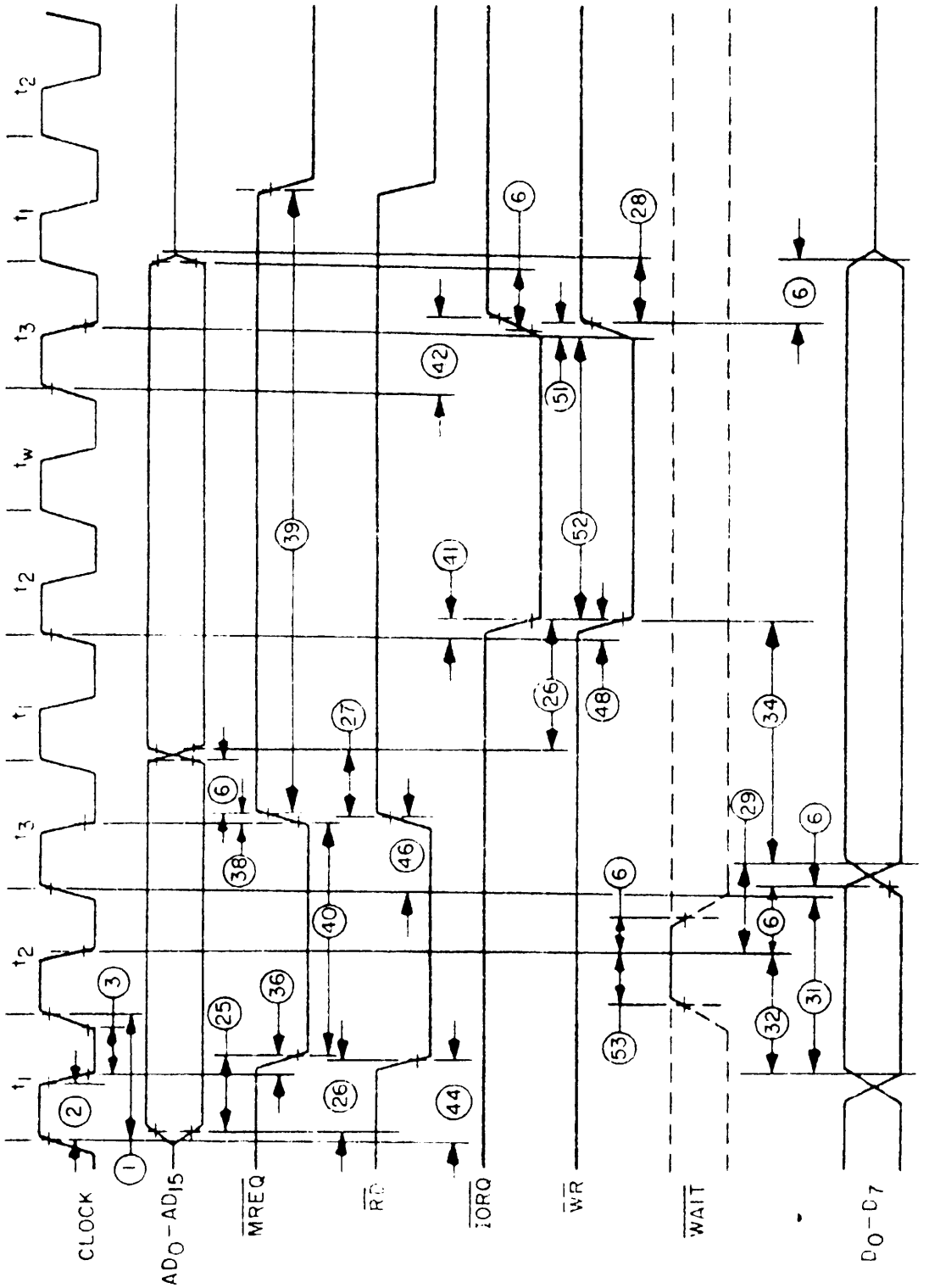


FIGURE 8. Memory read/IO write timing active state ac characteristics.

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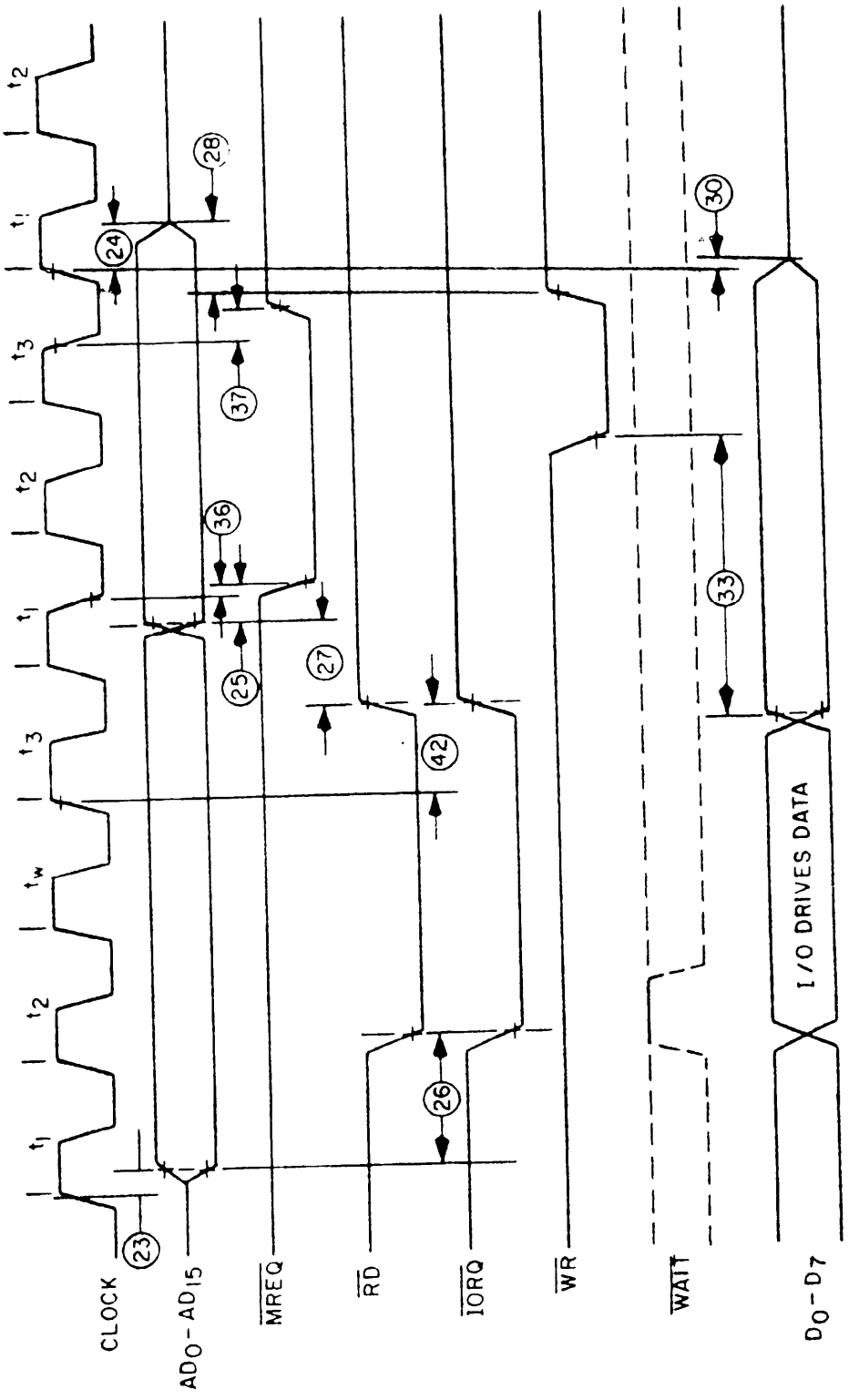


FIGURE 8. I/O read/memory write timing active state ac characteristics.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Constant acceleration in accordance with test condition B of method 2001 of MIL-STD-883.

d. The Percent Defective Allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroups 1 and 7, tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.7) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II, herein.

b. Subgroup 4 (C_{11}, C_{12}, C_O measurement) shall be performed only after process or design changes which may affect device capacitance.

c. Subgroups 5 and 6, table I of method 5005 of MIL-STD-883 shall be omitted.

d. Subgroup 12 shall be added to group A inspection requirements for class B devices using an LTPD of 10 and consist of the procedures, test conditions, and limits specified in table III. This subgroup shall be used for initial verification only and variables data shall be submitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883, and as follows:

A special subgroup shall be added to the group B inspection requirements for class B and shall consist of the test, conditions and limits specified in 4.5.2. The LTPD for this subgroup shall be 15 for all classes.

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4.4.3 Groups C and D inspections. Groups C and D inspections shall consist of the test subgroups and LTPD values shown in tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II, herein.

b. Steady-state life test (method 1005 of MIL-STD-883).

(1) Test condition D using the circuit shown on figure 3.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Constant acceleration in accordance with test condition B of method 2001 of MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 High-voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 400 V from a 100 pF source in the following test sequence.

a. Measure I_{IH} and I_{IL} at one input terminal of the device under tests at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

b. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in a 2-pulse sequence as follows and in accordance with the test circuit of figure 6.

(1) Input (-) to GND.

(2) Input (+) to GND.

c. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. In addition, all devices shall be in contact with a conductive material which shorts all leads together to prevent electrostatic damage.

6. NOTES

6.1 Test vector table. Table V, which forms a part of this specification, is not printed herein because of its extreme length and complexity; and, even when presented as hard copy, it requires conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be obtained in a form compatible with test system architecture (magnetic tape format usually preferred). The preparing activity (or DESC, as its agent) may be consulted for information on obtaining such output. For reference purposes, Rome Air Development Center (RADC-RBE-2) and Defense Electronics Supply Center (DESC-ECS) each maintain a limited quantity of dated, hard copy printouts of this table.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

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6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the data (see 4.4) pertinent to the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Terms and definitions. The abbreviations, terms, symbols, and definitions used herein (including terms and symbols for device terminals) are defined in MIL-M-38510, MIL-STD-1331, and as follows:

A_0-A_{15} (System address bus)	3-state output, active high. Addresses generated by the DMA are sent to both source and destination ports, either of which may be main memory or I/O peripherals, on these lines.
$\overline{BAI}$ (Bus acknowledge, IN)	Input, active low. Signals that the system buses have been released for DMA control. In multiple-DMA configurations, the $\overline{BAI}$ pin of the highest priority DMA is connected to the BUS Acknowledge pin of the CPU. Lower-priority DMAs have their $\overline{BAI}$ connected to the $\overline{BAO}$ of a higher-priority DMA.
$\overline{BAO}$ (Bus acknowledge out)	Output, active low. In multiple-DMA configurations, this pin signals that the CPU has relinquished control of the bus. $\overline{BAI}$ and $\overline{BAO}$ form a daisy chain for multiple-DMA priority resolution over bus control. Unlike the interrupt daisy chain formed with the IEI and IEO lines, this chain does not allow preemption of control by a high-priority DMA when a lower-priority DMA is already bus master. The DMA that has the bus is always allowed to finish, regardless of its priority in the chain.
$\overline{BUSRQ}$ (Bus request)	Bidirectional, active low, open drain. As an output it sends requests for control of the system address bus, data bus, and control bus to the CPU. As an input when multiple DMAs are strung together in a priority daisy chain via $\overline{BAI}$ and $\overline{BAO}$, it senses when another DMA has requested the buses and causes this DMA to refrain from bus requesting until the other DMA is finished. Because it is a bidirectional pin that allows simultaneous bidirectional signals with no means of control, there cannot be any buffers between this DMA and any other DMA. It can, however, have a buffer between it and the CPU because it is unidirectional into the CPU. A 1.8 kΩ pullup resistor is typically connected to this pin.

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$\overline{CE}/\overline{WAIT}$
(Chip enable and wait)

Input, active low. Normally this functions only as a $\overline{CE}$ line, but it can also be programmed to serve a $\overline{WAIT}$ function. As a $\overline{CE}$ line from the CPU, it becomes active when $\overline{WR}$ and $\overline{IORQ}$ are active and the I/O port address (up to 16 bits) on the system address bus is the DMA's address, thereby allowing control command bytes to be written from the CPU to the DMA. As a $\overline{WAIT}$ line from memory or I/O devices, after the DMA has received a bus acknowledge ($\overline{BUSAK}$) from the CPU, it causes wait states to be inserted in the DMA's operation cycles thereby slowing the DMA to a speed that matches the memory or I/O device.

CLK
(System clock)

Input, standard single-phase clock. For slower system clocks a TTL gate with a pullup resistor may be adequate to meet the timing and voltage level specifications. For higher-speed systems, use a clock driver with an active pullup to meet the V_{IH} specification and risetime requirements.

D_0 - D_7
(System data bus)

Bidirectional, 3-state. These pins transfer control bytes from the CPU, status bytes from the DMA, and data from memory or I/O peripherals. Data transfers or searches by the DMA are done only when the DMA controls this bus and the address bus. When the CPU controls these buses, it can write or read DMA control or status bytes.

IEI
(Interrupt enable in)

Input, active high. This line is used with IEO to form a priority daisy chain when there is no more than one interrupting device. A high on this line indicates that no other device of higher priority is interrupting, thereby allowing this DMA to interrupt if it is otherwise enabled to.

IEO
(Interrupt enable out)

Output, active high. IEO is high only if IEI is high and this DMA is not requesting an interrupt. Thus, this signal blocks lower-priority devices from interrupting while a higher-priority device is being serviced by its CPU interrupt service routine. Unlike devices in a bus-request daisy chain, devices in an interrupt daisy chain can be preempted by higher-priority devices before the lower-priority device has been fully serviced.

$\overline{INT}/\overline{PULSE}$
(Interrupt request)

Output, active low, open drain. This requests a CPU interrupt when brought low at a time when the DMA is not the bus master. The CPU acknowledges the interrupt by pulling its $\overline{IORQ}$ output low during $\overline{MI}$ cycle. The DMA $\overline{INT}$ pin is typically connected to the $\overline{INT}$ pin of the CPU with a pullup resistor and tied to all other INT pins in the system. This pin can also be used to generate periodic pulses to an external device. It can be used this way only when the DMA is bus master (i.e., the CPU's $\overline{BUSRQ}$ and $\overline{BUSAK}$ lines are both low and the CPU cannot see interrupts).

$\overline{IORQ}$
(Input/output request)

Bidirectional, active low, 3-state. As an input, this indicates that the lower half of the address bus holds a valid I/O port address for transfer of control or status bytes from or to the CPU, respectively; this DMA is the addressed port if its $\overline{CE}$ pin, $\overline{IORQ}$ pin, $\overline{WR}$ or $\overline{RD}$ pin are simultaneously active. As an output, after the DMA has taken control of the system buses, it indicates that the address bus holds a valid 16-bit port address for another I/O device involved in a DMA transfer of data. When $\overline{IORQ}$ and $\overline{MI}$ are both active inputs to the DMA simultaneously, an interrupt acknowledge by the CPU is indicated.

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$\overline{M1}$ (Machine cycle one)	Input, active low. This pin indicates that the current CPU machine cycle is an instruction fetch. It has two purposes in the DMA's interrupt structure. First, it is used by the DMA to detect return-from-interrupt instructions (RETI, or hex ED4D) fetched over the data bus by the CPU at the end of interrupt service routines. Second, an interrupt acknowledge is indicated when both $\overline{M1}$ and $\overline{IORQ}$ are active inputs to the DMA. During two-type instruction fetches, $\overline{M1}$ is active as each opcode byte is fetched.
$\overline{MREQ}$ (Memory request)	Output, active low, 3-state. This line indicates that the address bus holds a valid address for a memory read or write operation. After the DMA has taken control of the system buses, it indicates a DMA transfer request from or to memory.
$\overline{RD}$ (Read)	Bidirectional, active low, 3-state. As an input, this signal indicates that the CPU wants to read status bytes from the DMA's read registers, if selected. As an output, after the DMA has taken control of the system buses, it indicates a DMA-controlled read from memory or I/O port address.
RDY (Ready)	Input, programmable active low or high. This pin is monitored by the DMA to determine when a peripheral device associated with a DMA port is ready for a read or write operation. When the DMA is enabled to operate, the RDY line indirectly controls DMA activity; the manner in which DMA activity is controlled by RDY varies with the operating mode (Byte-Burst, or Continuous). An active RDY line can be simulated by programming a Force Ready condition. This is useful in memory-to-memory operations. It is preferable to have the RDY signal synchronized to the CLK signal, i.e., RDY should become active on the rising edge of CLK. This is particularly important in the Continuous mode of operation.
$\overline{WR}$ (Write)	Bidirectional, active low, 3-state. As an input, this indicates that the CPU wants to write control bytes to the DMA write registers when the DMA is selected. As an output, after the DMA has taken control of the system buses, it indicates a DMA-controlled write to a memory or I/O port address.

6.5 Logistic support. Lead material and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish "C" (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.6 Handling. MOS microprocessors should be handled with certain precautions to avoid damage due to the discharge of accumulated static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage, however, the following handling practices are recommended.

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

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6.7 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.3) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military-device type	Generic-industry type
01	Z80A DMA
02	Z80 DMA

6.9 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

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